

Capstone Project

Predicting Online Shopper Purchase Behavior Final Report

1. **Define the Problem Statement:** Online retail platforms attract many visitors, but most users leave without completing a purchase. This makes it difficult for businesses to know which users are likely to convert. The goal of this project is to build a machine learning solution that predicts whether a user will make a purchase during an online session.

Benefits:

- Improved personalization
- Increased conversion rates
- Better marketing efficiency

2. **Model Outcomes or Predictions:** This is a **supervised classification problem**.

Input: User session behaviour

Output: Purchase (Yes / No)

Both supervised and unsupervised methods were explored:

Supervised: Logistic Regression, Random Forest, SVM

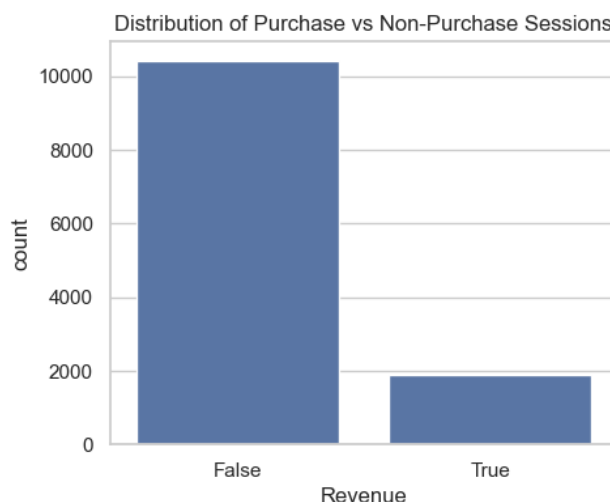
Unsupervised: PCA and K-Means clustering

3. **Data Acquisition:** The dataset was sourced from Kaggle and contains:

Page visits and durations

Traffic source and visitor type

Bounce and exit rates



The dataset is highly imbalanced, with only a small percentage of sessions resulting in a purchase. This imbalance reflects real-world online retail behavior but presents a challenge for machine learning models, as accuracy alone may give misleading results.

4. Data Preprocessing/Preparation: Data Preprocessing / Preparation

a. Cleaning Techniques

Checked for missing values
 Removed inconsistencies
 Converted categorical variables

b. Train-Test Split

80% training, 20% testing
 Stratified to preserve class balance

c. Encoding & Scaling

OneHotEncoder for categorical data
 StandardScaler for numeric features

5. Modelling: Multiple models were selected to compare different learning strategies:

Logistic Regression (baseline)
 Decision Tree
 KNN
 SVM
 Random Forest
 Each model was trained and evaluated using the same dataset.

6. Model Evaluation:

Due to class imbalance:

Accuracy alone was misleading
 Recall and F1-score were prioritized

...	Model	Accuracy	Precision	Recall	F1-Score
1	Decision Tree	0.893755	0.696078	0.557592	0.619186
4	Random Forest	0.897810	0.753906	0.505236	0.605016
3	SVM	0.888483	0.703422	0.484293	0.573643
0	Logistic Regression	0.881995	0.740741	0.366492	0.490368
2	KNN	0.868613	0.636792	0.353403	0.454545

Key Finding:

Logistic Regression achieved high accuracy but failed to identify buyers.
 Random Forest achieved better recall and overall performance.